# Usama Muhammad

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## Employment History

Oct 2025 – Present **AI Initiative Team Lead,** Braindeck Inc., Seocho-gu, Seoul, South Korea.

- Built a custom **ASR** model from scratch for Korean-language users with dysarthria, where no off-the-shelf solution was adequate, and integrated it into a real-time avatar communication platform with a **RAG**-based personalized communication tool.
- Fine-tuned and deployed **OpenAI Whisper** on custom datasets for domain-specific vocabulary and accent adaptation, and built **Voice Cloning** pipelines using **Fish Audio** (ASR, **TTS**, cloning) for on-premise deployment on limited GPU resources.
- Built an end-to-end pipeline for medically certified diagnosis, rehabilitation, and communication assistance for Korean-language users with dysarthria.
- Engineered the entire backend infrastructure for the avatar and medically certified rehabilitation applications utilizing **FastAPI**, and deployed these comprehensive **AI** services across **AWS**, **GCP**, and **Microsoft Azure**.
- Designed an advanced Multi-sensory Emotion Recognition Solution utilizing multisensor fusion (speech, facial expressions, biosignals) for precise emotion prediction.
- Architected a two-stage cascade **ML** system (**LightGBM** + **Random Forest**) for acoustic **NDT** defect classification of industrial fasteners (KPF), replacing a rule-based quality-control system with a data-driven **smart-manufacturing** pipeline over frequency-response (FRF) spectra.

Jul 2025 – Sep 2026 **AI Lead,** Brain Box Automations, Pakistan (Remote).

- Managed multiple projects in the AI department, supervising developers and maintaining client relations.
- Implemented **CI/CD for ML** pipelines and extensively developed production-grade AI agent-based services using **FastAPI**, **LangGraph**, and **LangChain** with focus on **LLMOps** and **MLOps**.
- Deployed scalable AI solutions on **AWS**, **GCP**, and **Microsoft Azure** platforms using **Docker** and **Microservices**.
- Built multi-agent research systems, niche coaching and cognitive assistants, AI-powered content management, and end-to-end recruitment automation pipelines with custom AI models, MCPs, and LLM tools, extensively leveraging **Gemini ADK** (Agent Development Kit) for complex multi-agent orchestration.

## Employment History (continued)

Aug 2017 – Aug 2026

**Graduate Student Researcher,** Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea.

- Developed advanced **Machine Learning** techniques for signal integrity and equalizer optimization (Samsung DS collaboration), applying latent representation learning for **Anomaly Detection** in DRAM signals.
- Led intelligent robotics research, developing robust visual odometry and localization methods for highly maneuverable robots utilizing **Deep Learning**.
- Built and deployed an end-to-end **ML** application for regulatory intelligence, extracting and summarizing safety and health policies from diverse government documents.
- Applied **Deep Reinforcement Learning** algorithms (**PPO**, **DDPG**, **Q-Learning**) to control systems including quadcopters and autonomous land robots.
- Engineered fair data representation methods to mitigate bias in downstream utility tasks, advancing transparency in **ML** models.

Aug 2023 – Sep 2025

**Freelance AI/ML Developer & Consultant,** Self-Employed (Remote).

- Developed Full-stack AI applications and multimodal platforms utilizing **React**, **Next.js**, and **GraphQL**.
- Engineered advanced AI agents with **OpenAI API** (**GPT-4o**), **Anthropic Claude API**, **Gemini API**, and open-source models (**Qwen**, **Mistral**) utilizing **RAG**, **Vector Databases** (Qdrant, **Pinecone**), and advanced **Context Engineering**.
- Built production **Voice AI** pipelines for KindredPM (US) using **Deepgram** real-time WebSocket **STT** with voice activity detection and **Aura TTS** for low-latency conversational agents, and integrated **Gemini Live API** for real-time speech-to-speech with barge-in support and multilingual dialog.
- Provided client consultation for **LLMOps** operations and production deployment.

Feb 2016 – Jul 2017

**Research Assistant,** NUST - Robotics & Intelligent Systems Engineering (RISE Lab), Islamabad, Pakistan.

- Developed wearable fNIRS device for real-time hemodynamic brain profiling.
- Implemented signal processing and ML techniques for brain signal feature extraction supporting motor neuron disability research.

Jul 2016 – Sep 2016

**Intern Application Engineer,** National Instruments (NI), Islamabad, Pakistan.

- Completed LabVIEW certification training.
- Designed and implemented "Hybrid Control of Inverted Pendulum" project on NI hardware platforms.

## Leadership, Entrepreneurship & Service

May 2026 – Present

**Founder / Tech Track Participant,** "Echo Project," Everyone's Startup Project (모두의 창업 프로젝트), Ministry of SMEs and Startups (Daejeon CCEI). Developing an **AI**-driven cognitive preservation platform; advanced via Fast-Track selection and secured **2,000,000 KRW** startup activity fund.

## Leadership, Entrepreneurship & Service (continued)

Aug 2025 – Nov 2025 — **Representative & Lead Developer,** Team 'Viro,' 2025 KAIST Startup Crew (Laboratory-Specialized Startup Leading University Program), KAIST. Directed technical architecture and application development; secured **2,000,000 KRW** in startup support funding.

Aug 2023 – Feb 2024 — **Part-time Researcher,** KAIST (Part-time Scholarship). Executed targeted research tasks under the supervision of Prof. Sang-Gug Lee and Prof. Thomas Cho.

2018, 2022 — **KAIST EE Ambassador,** EEIO Ambassador Program. Represented KAIST EE at the National University of Sciences and Technology (NUST), Pakistan, conducting multi-campus recruitment seminars for prospective graduate students.

2021, 2024 — **Short-term Teaching Assistant,** KAIST EE Visit Camp. Coordinated camp logistics and facilitated research lab tours for prospective international students.

Sep 2018 – Feb 2019 — **Teaching Assistant,** KAIST EEIO. Managed information desk operations, supported international students administratively, and organized EEIO Visit Camp events.

## Education

Aug 2019 – Aug 2026 — **Ph.D., Electrical Engineering,** Korea Advanced Institute of Science and Technology (KAIST).
GPA: 3.95/4.3 (Top 5%)
Thesis: *Learning Latent Representations for Anomaly Detection and Reinforcement Learning-Based Equalizer Optimization in High-Speed DRAM Systems*
Advisor: Prof. Dong Eui Chang

Aug 2017 – Aug 2019 — **M.S., Electrical Engineering,** Korea Advanced Institute of Science and Technology (KAIST).
GPA: 3.97/4.3 (Top 3.7%)
Thesis: *Towards Robust Neural Networks and Efficient Exploration in Reinforcement Learning*
Advisor: Prof. Dong Eui Chang

Sep 2013 – Jun 2017 — **B.E., Electrical Engineering,** National University of Sciences and Technology (NUST), Pakistan.
GPA: 3.99/4.0 (**Gold Medalist**)
Thesis: *Development of Functional Near Infrared Spectroscopy (fNIRS) based System for Detection of Distinctive Brain Signal Patterns*
Advisor: Prof. Fahad Mumtaz Malik
Co-Advisor: Prof. Yasar Ayaz
Awarded President of Pakistan Gold Medal for Best Graduate in Electrical Engineering

## Research Publications

### Patents

1. **M. Usama** and D. E. Chang, "Reinforcement learning-based equalizer parameter optimization method and system (original title: 강화학습 기반 이퀄라이저 파라미터 최적화 방법 및 시스템)," Korea Patent Application 10-2025-0119661, Aug. 2025.

2. **M. Usama** and Y. R. Jung, "Method and apparatus for monitoring health status of patient based on reliability using multiple biosignals (original title: 복수의 생체 신호를 이용하여 신뢰도를 기반으로 환자의 건강 상태를 모니터링하는 방법 및 장치)," Korea Patent Application 10-2025-0213058, Dec. 2025.

## Journal Articles (Accepted / Published)

1. E. E. Pazo, X. Liu, S. Liu, Q. Zhang, M. Zhang, Z. Zhang, S. Moutari, **M. Usama**, and X. Ren, "Validation of the Eyerobo FC portable fundus camera for diabetic retinopathy screening using public datasets and deep learning," *Ophthalmology and Therapy*, 2026. DOI: 10.1007/s40123-026-01353-w.

2. H. U. K. Shinwari and **M. Usama**, "Memory-augmented architecture for long-term context handling in large language models," *IEEE Transactions on Artificial Intelligence*, 2026, Equal contribution. DOI: 10.1109/TAI.2026.3701680.

3. **M. Usama**, "The GRPO tax is smaller than you think: A longitudinal study of capability preservation during reasoning training," *Transactions on Machine Learning Research (TMLR)*, 2026. URL: https://openreview.net/forum?id=e0UVcimXdK, Code: https://github.com/Usama1002/GRPO-Capability-Tax.

4. **M. Usama** and D. E. Chang, "Distributional reinforcement learning with information bottleneck for uncertainty-aware DRAM equalization," *IEEE Transactions on Components, Packaging and Manufacturing Technology*, vol. 16, no. 8, pp. 1912–1929, 2026. DOI: 10.1109/TCPMT.2026.3674506.

5. **M. Usama** and D. E. Chang, "Multi-lane joint DRAM equalization via cooperative multi-agent distributional reinforcement learning," *IEEE Transactions on Components, Packaging and Manufacturing Technology*, 2026. DOI: 10.1109/TCPMT.2026.3711630.

6. **M. Usama** and Y. Jung, "Evidential multimodal transformers with Dempster-Shafer fusion for emotion recognition," *IEEE Transactions on Affective Computing*, 2026. DOI: 10.1109/TAFFC.2026.3742820.

7. N. Wei, C. Li, S. Moutari, **M. Usama**, J. Li, Q. Chen, E. E. Pazo, and X. Qian, "Predicting pediatric myopia progression from single screening encounters: A multi-outcome development and validation study," *Journal of Translational Medicine*, 2026. DOI: 10.1186/s12967-026-09003-2.

8. M. Xia, B. Yao, Y. Wang, F. Liang, W. Zhao, L. Qin, S. Liu, **M. Usama**, Z. Zhang, W. Guo, E. E. Pazo, and X. Ren, "Machine learning-based prediction of cycloplegic refraction using the Eyerobo vision screener: Design—a cross-sectional comparative device study," *Ophthalmology and Therapy*, 2026. DOI: 10.1007/s40123-026-01482-2.

9. E. E. Pazo, S. Moutari, F. Gao, L. Hu, **M. Usama**, X. Li, and J. Liu, "Structure-preserving super-resolution of retinal fundus images via a dual-transformer residual network," *Frontiers in Medicine (Section: Ophthalmology)*, 2025. DOI: 10.3389/fmed.2025.1730678.

10. H. U. K. Shinwari and **M. Usama**, "ARD-LoRA: Dynamic rank allocation for parameter-efficient fine-tuning of foundation models with heterogeneous adaptation needs," *IEEE Transactions on Artificial Intelligence*, 2025, Equal contribution. DOI: 10.1109/TAI.2025.3605569.

11. **M. Usama**, H.-D. Jang, and D. E. Chang, "Deep reinforcement learning-based DFE structure optimization for high-speed DRAM signals," *IEEE Transactions on Components, Packaging and Manufacturing Technology*, 2025. DOI: 10.1109/TCPMT.2025.3648009.

12. **M. Usama** and D. E. Chang, "Learning flexible and fair data representations," *IEEE Access*, vol. 10, 2022. DOI: 10.1109/ACCESS.2022.3207747.

## Journal Articles (Under Review)

1. **M. Usama**, "Delta-CKA and the warm-start imperative: Empirical findings on layer compression in large language models," *Transactions on Machine Learning Research (TMLR)*, 2026, Under Review.

2. **M. Usama** and D. E. Chang, "Convergence without understanding: When language models agree on representations but disagree on reasoning," *Journal of Machine Learning Research (JMLR)*, 2026, Under Review. arXiv: 2605.23315.

3. **M. Usama** and D. E. Chang, "The system prompt illusion: How instruction preambles modify computation in language models," *Journal of Machine Learning Research (JMLR)*, 2026, Under Review. arXiv: 2609.38205.

4. **M. Usama** and D. E. Chang, "Deep reinforcement learning-based DRAM equalizer parameter optimization using latent representations," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2025, Under Review.

5. **M. Usama**, H.-D. Jang, S. Shanbhag, Y.-C. Sung, S.-J. Bae, and D. E. Chang, "Learning high-quality latent representations for anomaly detection and signal integrity enhancement in high-speed signals," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2025, Under Review.

### Conference Proceedings

1. **M. Usama** and Y. Jung, "Beyond adjacent layers: Graph-guided layer fusion for compressing large language models," in *Conference on Neural Information Processing Systems (NeurIPS)*, Accepted, 2026. arXiv: `2610.15159`, Code: `https://github.com/Usama1002/fcm-layer-fusion`.

2. **M. Usama** and Y. Jung, "F-ConvNeXt: Frequency-aware domain generalization for deepfake detection via self-blended synthetic training," in *Conference on Neural Information Processing Systems (NeurIPS)*, Under Review, 2026.

3. **M. Usama** and D. E. Chang, "Uncertainty-aware DRAM equalizer optimization," in *2025 Summer Annual Conference of the Korean Institute of Electromagnetic Engineering and Science (KIEES)*, 2025.

4. **M. Usama** and D. E. Chang, "Uncertainty-aware DRAM equalizer optimization using Bayesian representation learning and deployment risk assessment," in *2025 Conference on Electronics, Semiconductors, and Artificial Intelligence (CESAI)*, 2025.

5. **M. Usama** and D. E. Chang, "Optimization of decision feedback equalizer structure through genetic algorithm and neural network evaluation," in *39th Institute of Control, Robotics and Systems (ICROS) Conference 2024*, 2024.

6. **M. Usama** and D. E. Chang, "Particle swarm optimization for DFE structure optimization with neural fitness evaluation," in *2024 Summer Annual Conference of IEIE*, 2024.

7. **M. Usama** and D. E. Chang, "Representation learning for distortion classification in DRAM data," in *39th Institute of Control, Robotics and Systems (ICROS) Conference 2024*, 2024.

8. **M. Usama** and D. E. Chang, "PyEye: An integrated approach for signal integrity assessment and eye diagram generation," in *2023 IEEE International Conference on Consumer Electronics-Asia (ICCE-Asia)*, 2023.

9. **M. Usama** and D. E. Chang, "Learning-based control of quadrotor using Gaussian processes," in *6th Joint Conference of Korean Artificial Intelligence Association (CKAIA 2022)*, Recipient of Special Society Award, 2022.

10. **M. Usama** and D. E. Chang, "Learning-driven exploration for reinforcement learning," in *2021 21st International Conference on Control, Automation and Systems (ICCAS)*, 2021.

11. D. Lee, **M. Usama**, and D. E. Chang, "Control of Furuta pendulum with reinforcement learning," in *19th International Conference on Control, Automation, and Systems (ICCAS 2019)*, 2019.

12. **M. Usama** and D. E. Chang, "Robotic navigation using entropy-based exploration," in *2019 19th International Conference on Control, Automation and Systems (ICCAS)*, 2019.

13. **M. Usama** and D. E. Chang, "Towards robust neural networks with Lipschitz continuity," in *17th International Workshop on Digital Forensics and Watermarking (IWDW 2018)*, 2018.

## Skills

AI/ML — Deep Learning, Reinforcement Learning, Large Language Models (LLMs), Multi-Agent Systems (**LangGraph**, **Cognitive Architectures**), Foundation Model Fine-tuning, Parameter-Efficient Training (LoRA, ARD-LoRA), Computer Vision, NLP, **HuggingFace Transformers**, **Qwen**, **Mistral**, **LightGBM**, Gradient-Boosted Trees, **PPO**, **DDPG**.

## Skills (continued)

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| LLM Ops | Context Engineering (**ReAct Context Engineering**), RAG Systems, Vector Databases, **MCP Server Development** (published integrations for ChatGPT, Claude, Perplexity), Custom LLM Tools, **OpenAI API**, **Anthropic Claude API**, **Gemini API**, **Gemini ADK**, Production LLM Deployment, **LLMOps**, **MLOps**, **NL2SQL**. |
| Voice AI | **Deepgram** (Real-time STT/TTS), **OpenAI Whisper** (Fine-tuning), **Fish Audio** (ASR/TTS/Voice Cloning), **Gemini Live API** (Speech-to-Speech), Custom ASR Training, Voice Activity Detection, Low-latency Conversational AI. |
| Coding | Python, PyTorch, TensorFlow, TypeScript, JavaScript, React, **Next.js**, Node.js, FastAPI, Flask, **Django**, **GraphQL**, LaTeX. |
| Tools | Git, **Docker**, **CI/CD**, Linux, LabVIEW, MATLAB, Gradio, Streamlit, Jupyter, VS Code. |
| Cloud & DB | **AWS**, **GCP** (Vertex AI, Cloud Scheduler), **Azure**, PostgreSQL, **pgvector**, **Neo4j**, **Elasticsearch**, SQLite, MongoDB, Qdrant (Vector DB), **Pinecone**, Redis. |
| Research | Academic writing, peer-reviewed publication, experimental design, statistical analysis, IEEE and journal submissions. |
| Languages | English (Fluent), Urdu (Native), Korean (Intermediate – TOPIK Level 2). |

## Honors & Awards

### Awards and Achievements

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| 2026 | **Gold Reviewer**, ICML 2026 (International Conference on Machine Learning). Top reviewer recognition based on area chair ratings; awarded complimentary conference registration. |
| 2025 | **Winner – Logo Design Contest**, KAIST Space Institute. Worth 1,000,000 KRW. |
| | **Excellent Idea Award**, AI Technology Utilization in KAIST Administration Idea Contest, KAIST Center for Administration Development. Worth 300,000 KRW. |
| 2017 | **President of Pakistan Gold Medal**, Best Graduate in Bachelor of Engineering, NUST. |
| 2022 | **Special Society Award**, Worth 1 million KRW, Korean Artificial Intelligence Association. |
| 2019 | **AlphaPilot Qualifier**, Lockheed Martin AI Drone Racing Innovation Challenge. Team Puffin ranked 7th among 400+ teams. |
| 2017 | **Runners Up**, All Pakistan Engineering Category, AirTECH 2016, Air University. |
| | **Runners Up**, All Pakistan Embedded Systems Category, COMPPEC'17, CEME NUST. |
| 2013–2016 | **EME High Achievers Awards**, Consecutive recognition for academic excellence. |
| 2013 | **5th Position**, Among 102,080 students in FBISE HSSC Examination. |

### Certification

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| 2016 | **Certified Associate Developer – LabVIEW**. Awarded by National Instruments. |
| | **PLC, HMI and SCADA Systems**. Professional Training. |

## References

Available on Request